

Problem Statement

Objective: Build an independent, deviation-aware encoding layer. By structuring behavioural changes at the entity level before detection, we successfully separate data representation quality from the choice of detector. Cloud systems now include IoT devices, microservices, edge nodes, and automated agents. Detectors exist (Isolation Forest, autoencoders, density-based methods), but they operate on **raw telemetry**, where low-amplitude, multi-day drift looks indistinguishable from normal variation

- Issue 1:** Existing UBA frameworks are built around human habits like typical 9-to-5 work hours or standard session lengths, which simply don't apply to automated bots or IoT devices.
- Issue 2:** Traditional point-in-time scoring methods (like LOF) fail to catch slow, coordinated attacks that subtly shift across multiple dimensions over several days.
- Issue 3:** Right now, there isn't a framework that untangles the structuring of data deviations from the actual detection process to create a truly independent, reusable layer.

Dataset & Protocol

Dataset: 50 entities across 8 zones and 7 classes over 59 periods.
Anomalies: 5 categories (temporal, topology, volumetric, privilege, state).
Protocol: Chronological 31/31 train/test split prevents data leakage.

MAD-EBA Framework Summary

Partition Data: Group incoming raw telemetry by entity class and deployment zone.
Calculate Deviation: Compare feature deviations against a rolling peer baseline and the entity's 30-period history.
Apply Thresholds: Use a per-feature threshold θ_f that balances a strict operational minimum with dynamic peer percentiles.
Generate Spike Vectors: Aggregate weighted spikes into a single vector ($v_{e,t}$), capping minor dimensional impacts at 0.18.
Feed Detectors: Pass the rich spike vector ($v_{e,t}$) rather than a basic binary flag to downstream models like IF, Deep AE, or MLP-AE.
Escalate Smartly: Trigger alerts only when deviation magnitude, peer divergence, and temporal persistence strongly align.

MAD-EBA Pipeline



Research Gaps & Contributions

Issues with existing approaches:

- Missing encoding layer:** Detectors process data before deviations are structured against peer and historical baselines.
- No temporal persistence:** Current systems treat brief, harmless noise and sustained, coordinated attacks exactly the same.
- Tangled architecture:** Existing methods fail to separate deviation structuring into an independent, reusable layer.

MAD-EBA's three contributions:

- Deviation-as-representation:** Spike vectors explicitly encode anomalies without needing training data for the encoding step.
- Entity-native baselines:** Profiles are built on device class and deployment zone, avoiding flawed human-centric assumptions.
- Detector-agnostic validation:** Our generated spike vectors improve F1 scores across three completely different model families.

Deviation Structuring & Seven Windows

Per-feature threshold combines a static floor with a dynamic peer norm:

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$$\theta_f = \max( pf, 0.3 \cdot pf + 0.7 \cdot \beta \cdot P95cz )$$

$$\beta = 1.2 - \text{grid-searched over } \{1.0, 1.1, 1.2, 1.3, 1.5\}$$

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Features are grouped into seven operational windows: connection activity, peer topology, operational state, data operations, data transfer, service interactions and resource consumption.
An alert escalates only when **multiple windows** confirm coordinated deviation, preventing single-window trigger inflation.

Limitations & Analysis

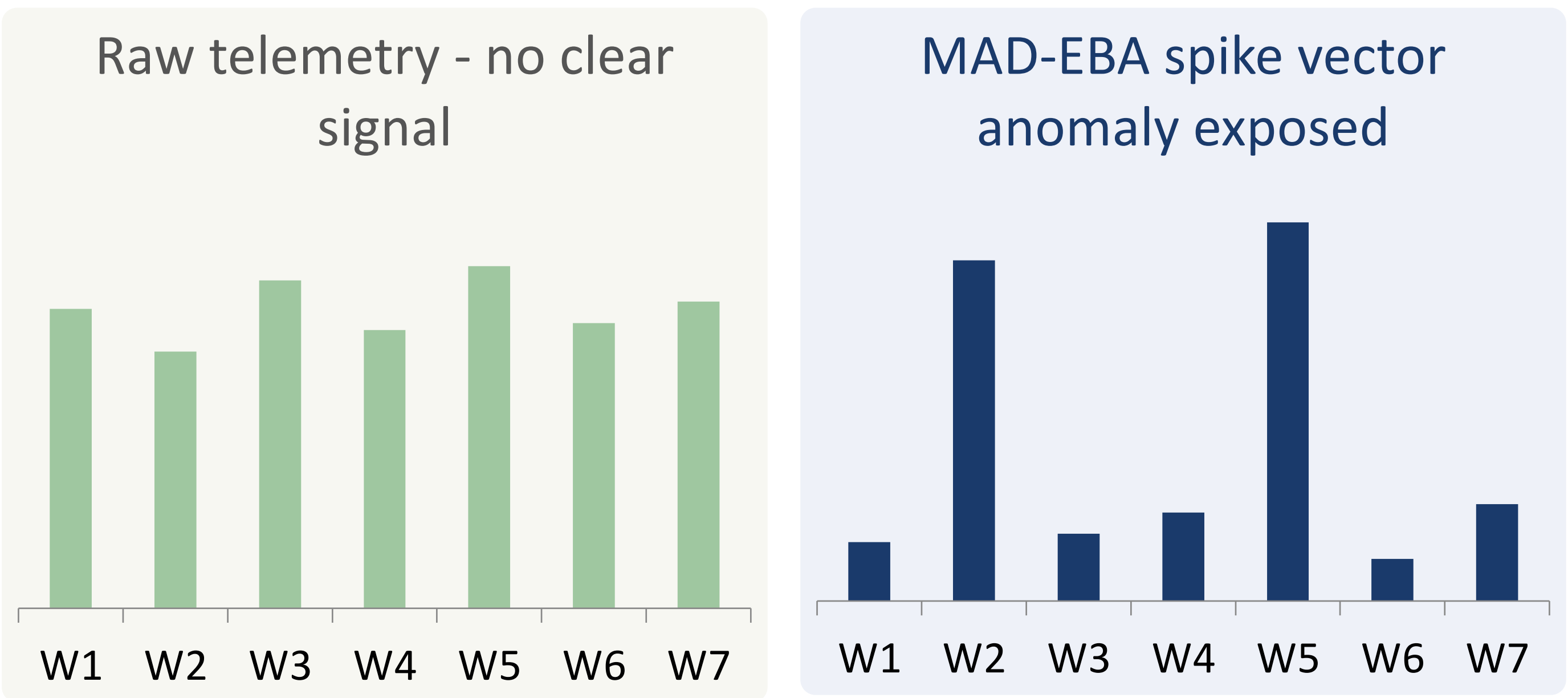
Hardest to detect: Persistent state anomalies and slow topology shifts are challenging because their deviations hover right at the threshold boundary, resulting in low-amplitude spikes.

Easiest to detect: Privilege and configuration changes are caught highly reliably, as they trigger sharp spikes across multiple operational windows.

Caveats: Our current analysis relies on a controlled dataset with a small population of 50 entities. We have not yet evaluated the framework's robustness against adversarial attacks or highly noisy real-world telemetry

Results

0.774		+9.5%		2,425
BEST F1 (MLP-AE)		GAIN VS. BASELINE		ENTITY-PERIOD PROFILES
Config	Detector	Prec.	Rec.	F1
Raw features	IF	0.697	0.719	0.707
MAD-EBA	IF	0.738	0.797	0.762 +5.5%
MAD-EBA	Deep AE	0.762	0.781	0.769
MAD-EBA	MLP-AE	0.759	0.789	0.774 +9.5%



Same entity-period profile raw telemetry (left) vs. MAD-EBA spike vector (right). W2 (topology) and W5 (transfer) expose the coordinated anomaly.

LOF: ROC-AUC = 0.041 on normalised spike vectors — reconstruction-based methods (AE, MLP-AE) are the appropriate comparators.

Future Direction

- Scale validation** Testing on HDFS telemetry (11 million events, 203 nodes) to target a 100x scale increase.
- Extended detectors:** Pairing our generated spike vectors with LSTM-AE and graph-based anomaly detectors
- Dynamic roles:** Reassigning peer groups on the fly as edge nodes alternate duties, completely bypassing the need to retrain the baseline.
- Pipeline integration:** Embedding seamlessly into existing SIEM or cloud monitoring platforms without altering the detection stage.

References

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